FINAL INTERNSHIP REPORT

AI CHATBOT DEVELOPMENT FOR CAVALONS

CESI_
ÉCOLE D'INGÉNIEURS

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Name of the class: A3 in computer science engineering for English speakers

This document has been written in Toulouse, France, on March 31, 2026

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The full internship period: from 12 January to 8 March 2026

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1. Introduction

1.1. Report Overview and Objectives

This document constitutes the final internship report for me, Manil Doudou, a third-year student in the CESI *diplôme d'ingénieur* program for English speakers at CESI Toulouse. It provides a comprehensive and detailed account of the ten-week internship undertaken at Cavalons, a French startup based in Paris operating in the equestrian technology sector, from January 12 to March 8, 2026. The primary objective of this report is to fulfil the academic requirements of the A3 internship. This involves detailing the professional mission, the context in which it was carried out, the methodologies employed, the technical work completed, and a critical analysis of the outcomes. Furthermore, the report serves as a personal and professional reflection on the integration of theoretical knowledge acquired during the engineering curriculum into a practical, real-world setting. It aims to demonstrate the acquisition of technical mastery, the development of professional competencies, and an understanding of the complex dynamics within an innovative startup in France in 2026.

1.2. Personal and Professional Motivations

My motivation for pursuing this specific internship was at its peak since the beginning of the internship research process. Firstly, I was eager to apply and deepen my knowledge in the rapidly evolving field of **Artificial Intelligence**, particularly in **Conversational AI and Natural Language Processing (NLP)**. The opportunity to work on the development of an AI-powered coaching assistant from its conceptual stages was an ideal challenge to bridge my academic learning with cutting-edge industry practice. Secondly, I was drawn to the startup environment at Cavalons. I sought an immersive experience where I could witness the entire product development lifecycle, from understanding market needs to technical implementation and strategic pivots. The chance to work closely with a small, agile team, taking on significant responsibility and contributing directly to a concrete product, was a powerful motivator. The unique domain (Equestrian world) added an appealing layer of complexity, requiring me to quickly acquire domain-specific knowledge and apply my engineering skills to a new and meaningful context.

1.3. Report Structure

This report follows the internship journey chronologically and thematically. Section 2 presents Cavalons, its market, and strategic vision, while Section 3 defines the context and objectives of the ÉquiProgress project. The work then unfolds across four phases: Section 4 covers the initial discovery, onboarding, and early data analysis; Section 5 dives into the deep technical development of the chatbot's core architecture; Section 6 addresses the integration challenges encountered with the application's backend; and Section 7 focuses on strategic research and theoretical consolidation. Section 8 synthesizes the final deliverables and Section 9 details the project management methodologies employed. Sections 10 and 11 offer a critical reflection on skills acquired and challenges faced alongside the solutions implemented, while Section 12 provides strategic observations and recommendations. The report closes with a personal reflection in Section 13, acknowledgements in Section 14, and supplementary materials like a bibliography in Section 15 and finally some appendices in Section 16.

2. Company Presentation: Cavalons

Figure 1: An example of one of the banners of the company's website www.cavalons.fr

2.1. Genesis and Core Business Model

Cavalons is a French startup founded in 2023 by Camille Despesse (CEO) and Théo Cousinet (Lead Developer). The company was established to address significant barriers to entry in the equestrian world, namely high costs and lack of accessibility. The traditional model often requires individuals to own a horse or be a long-term member of an expensive equestrian centre, which excludes a large portion of potential enthusiasts.

Cavalons' initial business model is a digital platform that facilitates a peer-to-peer sharing economy within the equestrian community. Its core services are designed to connect horse owners with riders who do not own a horse:

- **Demi-pension (Part-Lease):** A structured, long-term arrangement where a rider uses a horse several times a week in exchange for a fee and sometimes help with care. This provides the owner with financial help and the rider with regular access to a horse.
- **Horse Sitting:** It's like baby-sitting but for horses, it's a short-term care service, akin to pet sitting, where a rider looks after a horse while the owner is away, ensuring the animal's routine and well-being are maintained.

- **Experience Sharing:** A service for one-off or occasional rides, lessons, or trail rides, allowing users to discover new horses, disciplines, or simply enjoy a ride without any specific commitment.

This platform model leverages technology to build trust, optimize the use of equine resources, and significantly lower the financial threshold for participating in equestrian activities (for example: Dressage, Show jumping, Endurance riding, etc).

2.2. Market Analysis: The French Equestrian Sector

The company operates within a substantial and deeply rooted market. The French equestrian sector is one of the largest in Europe, comprising approximately **2.5 million equids and 2.7 million active riders**. It generates an estimated **€11 billion in annual economic activity**, encompassing breeding, training, competitions and equipment.

Despite this scale, the sector has been traditionally structured around physical infrastructures (equestrian centres) and ownership. Digital transformation has been slow, creating a ripe opportunity for tech-enabled services. Cavalons positions itself at the intersection of the sharing economy specialized in leisure services. By digitizing connections and building a community, Cavalons taps into a growing demand for flexibility, accessibility, and trust-based transactions in this specific domain.

2.3. Strategic Evolution: From Platform to Ecosystem

During my internship, it became clear that Cavalons is not resting on its initial success. The company is undergoing a significant strategic evolution with the development of its new application, **ÉquiProgress** (later rebranded as "Jappeloup" during the internship). This initiative marks a transition from a purely connective platform to a comprehensive digital ecosystem. The vision for ÉquiProgress is to become the essential digital companion for any equestrian. While the original platform solved the problem of *access* ("find a horse to ride"), ÉquiProgress aims to solve the problems of *quality of experience* ("how do I train effectively and safely with this horse?"). It is conceived as a a full ecosystem including an **AI-powered coaching assistant** that will provide:

- **Personalized Session Planning:** A structured intake process collects key information on the horse and rider before generating tailored, progressive, and safe training sessions.

- **Structured Training Delivery:** Sessions are presented as clear, chronological to-do lists with justifications and warning signals for each exercise.

- **Equine Wellness & Workload Monitoring:** Health data, veterinary visits, and past workload are logged and systematically factored into every recommendation, keeping horse welfare at the centre.

- **Multi-User Coordination:** Owners, part-leasers, and coaches share a centralized data source to ensure a consistent and safe workload.
- **Long-Term Progress Tracking:** Sessions are linked to past work and future goals, with post-session questionnaires and calendar logging to visualize progression.
- **Scoped Equestrian Expertise:** ÉquiProgress strictly focuses on equestrian topics, ensuring professional and responsible guidance at every interaction.

This product extension deepens user engagement, transforms a transactional relationship into a long-term, value-added partnership, and unlocks new revenue streams (e.g., premium subscriptions), significantly increasing the company's market valuation and strategic importance while creating a new dynamic.

2.4. Organizational Structure and Team Dynamics

Cavalons maintains a lean and agile operational structure, characteristic of an early-stage startup. The core team consists of the two founders:

- **Camille Despesse (CEO):** Responsible for strategic direction, product vision, market analysis, user research, and business development. She acts as the product owner, translating market needs into clear requirements.
- **Théo Cousinet (Lead Developer):** Oversees the entire technical architecture, frontend and backend development (Supabase, cloud functions), and ensures system scalability, performance, and security.

My role as an **A3 intern** was integrated as a crucial third member of this core team for the duration of the internship. This structure fostered incredibly flat hierarchies and direct communication. Decisions could be made rapidly in conversations between the three of us, and I was entrusted with significant responsibility from day one. The dynamic was highly collaborative, with knowledge flowing freely in all directions.

2.5. Company Culture: Values of Sharing, Trust, and Safety

The company's stated values are not mere platitudes but are actively embodied in daily operations, as I experienced firsthand and I can say that they are properly applied.

- **Sharing:** This is the bedrock of both the product and the internal team culture. Knowledge, ideas, and challenges are shared openly. My daily stand-ups were a forum for collective problem-solving, not just status updates. The founders were incredibly generous with their time and expertise.

- **Trust:** This is the foundation of the working relationship. I was given a high degree of autonomy and ownership over my tasks. This trust was reciprocated through transparency about the company's challenges, roadmap, and strategic decisions. It created a psychologically safe environment where asking questions and admitting uncertainty was encouraged.
- **Safety:** This is paramount, applied both to the product and the workplace. For the ÉquiProgress chatbot, safety translates into rigorous protocols to ensure AI recommendations never harm a horse or rider. Internally, it manifests as psychological safety, a workplace where one can take risks, propose unconventional ideas, and learn from mistakes without fear of negative judgment. This value was critical for my own development, allowing me to experiment and grow confidently.

3. Internship Context and Project Framework

3.1. Pedagogical Objectives of the A3 Internship

The third-year internship is the first in-company training period in the engineering cycle, designed to provide a suitable environment for acquiring technical know-how and applying skills developed during training. Its intended outcomes include producing technical work, identifying task challenges, organizing work within a professional environment, and summarizing results in both written and oral form, all while adopting a French professional behaviour and values. More specifically, the internship aims to develop an understanding of the company's organization, integrate into it meaningfully, and carry out a Bachelor-level assignment related to one's field of specialization (in this case, Computer Science with a focus on AI & Data Science) while deepening knowledge in relevant technical areas. This internship at Cavalons was perfectly aligned with these objectives through the development of the ÉquiProgress AI chatbot.

3.2. Project Origin: The Vision of an AI Coach

The ÉquiProgress AI Coach project was born from a deep understanding of the end-user's daily frustrations, or "pain points". The founders recognized that beyond finding a horse to ride, users struggled with:

- **Lack of Guidance:** Riders often face "blank page syndrome," not knowing what exercise to do in a session to progress towards their goals.
- **Lack of Transparency:** In a multi-user scenario (owner + part-leaser), it's difficult to know what work has been done, leading to fears of overworking or mistraining the horse.

- **Lack of Motivation:** Progress towards long-term goals can feel slow and invisible without a structured way to track it with databases and visuals.
- **Lack of Health Oversight:** Owners and riders can forget or miscommunicate about important health events (vet visits, farrier appointments, injuries).

The vision was to create an intelligent system that acts as a "digital coach" and a "central nervous system" for the horse's well-being, addressing all these points in a single, cohesive user experience.

3.3. Problem Statement: Addressing Key User Pain Points

The core problem the project sought to solve was the **fragmentation of information and lack of intelligent guidance** in the daily management and training of a shared horse. The goal was to create a solution that could:

- **Structure** training towards specific, measurable goals.
- **Minimize risks** of injury or under-performance caused by poor team coordination or lack of technical knowledge.
- **Motivate users** by making progress visible and providing a clear path forward.

This required moving beyond a simple FAQ chatbot to a context-aware system capable of synthesizing various data points: rider profile, horse profile, training history, health events, and upcoming appointments.

3.4. Primary Mission: Design and Integrate the Core AI Chatbot

My primary mission, as defined in my internship agreement and validated by my academic tutor, was to **design, develop, and integrate the core AI-powered coaching chatbot for the ÉquiProgress application.**

This overarching mission broke down into several key responsibilities:

1. **Conceptualization:** Define the chatbot's "personality," scope, and core functionalities based on the product specifications.
2. **Technical Development:** Build the conversational logic using a suitable platform (Botpress), integrating a Large Language Model (LLM) via API (Gemini).
3. **Knowledge Curation:** Gather, structure, and integrate domain-specific equestrian knowledge to ground the AI's recommendations and ensure their pedagogical soundness and safety according to the company's values.
4. **Integration:** Connect the chatbot to the application's backend (Supabase) to allow it to access real user data for hyper-personalization.

5. **Testing:** Implement a testing protocol to validate the chatbot's responses for safety, relevance, and user experience, iterating based on feedback.

3.5. Secondary Missions and Deliverables

In addition to the primary development mission, several secondary missions were integral to the internship that were more or less related to the main objective:

- **Data Analysis:** Analysing existing equestrian datasets (from Cavalons' network, Ekwo, Appaloo) to identify training patterns, common bottlenecks, and evidence-based recovery protocols to inform the chatbot's logic.
- **Backend Support:** Assisting the Lead Developer with database design and management in Supabase to prepare the backend for connection with the Figma mock-up and the chatbot knowledge base.
- **Strategic Research:** Conducting deep, theoretical research (like mini-courses showed on PowerPoint) on the latest advancements in AI and chatbot building technologies (RAG, multi-modality, LLM comparison) to provide the company with forward-looking strategic recommendations for their future projects.

The key deliverables were the functional chatbot on Botpress, the integrated prototype on the ÉquiProgress website mock-up, a structured knowledge base, a documented database schema, and the strategic research presentations.

4. Phase 1: Discovery and Onboarding (Weeks 1-3)

4.1. The Discovery Report: A Fresh Perspective on Cavalons

As mandated by CESI, the first major deliverable of the internship was the Discovery Report. This exercise was instrumental in my integration. Writing it forced me to take a step back from the daily technical immersion and analyse Cavalons from a structural, cultural, and strategic perspective. As a newcomer, I was able to provide an outside perspective on the company's operations. I observed the lean structure, the agile workflows, and the strong embodiment of corporate values. The report highlighted Cavalons' strengths: a clear strategic vision, a highly collaborative culture, and a focused, mission-driven team. It also allowed me to formulate initial observations about potential future challenges, such as knowledge preservation as the team scales, which I later expanded upon in my final recommendations. This exercise was not just an academic formality; it was a valuable reflective practice that primed me for deeper engagement with the present and future company's challenges.

4.2. Initial Integration and Onboarding Process

The onboarding at Cavalons was remarkably efficient and immersive. On the very first day, I received a comprehensive dual briefing including all the members of Cavalons:

- **Strategic Onboarding with Camille (CEO):** We discussed the company's founding story, a detailed market analysis of the French equestrian sector, the competitive landscape, and the strategic roadmap for ÉquiProgress. This provided the crucial "why" behind the project.
- **Technical Onboarding with Théo (Lead Developer):** This session was a deep dive into the existing and planned technology stack. Théo introduced me to React Native for the frontend, Supabase as the backend-as-a-service, and the architecture for the new ÉquiProgress project. We set up my local development environment, and I was granted access to all relevant repositories, documentation, and project management tools (Trello).

4.3. Understanding the ÉquiProgress Ecosystem

A significant part of the first week was dedicated to deeply understanding the ÉquiProgress ecosystem. I studied the initial product specifications, the user stories, and the detailed technical requirements outlined in the different documents given.

I familiarized myself with the core components of the envisioned AI Coach:

- **The Recommendation Engine:** Its need to process rider objectives, session history, and horse health data to give the best recommendation possible.
- **Conversational Interaction:** The requirement for a natural chat interface.
- **Proactive Capabilities:** The AI's ability to initiate conversations for feedback.

This foundational understanding was critical for the technical design work that would follow. I also began researching the equestrian domain, learning key terminology and training principles to ensure I could credibly contribute to a product in this niche field.

4.4. First Technical Contributions: Data Analysis

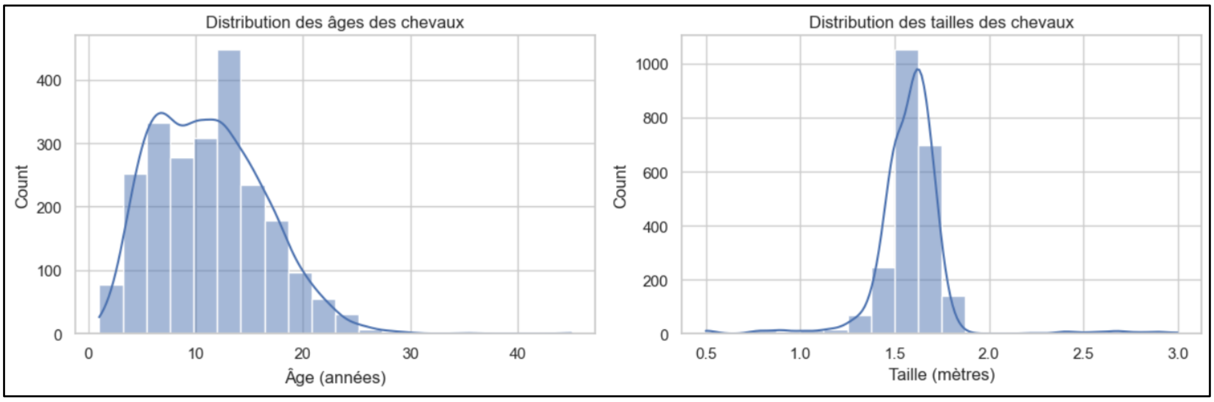


Figure 2: Data visualization of the age distribution of horses & the height distribution of horses

Even during this initial discovery phase, I began contributing technically. My first assignments involved data analysis, which served a dual purpose: it provided immediate value to the company and deepened my own understanding of the domain.

4.4.1. Analysing Cavalons, Ekwo and Appaloo Datasets

I was given anonymized datasets from Cavalons and from other equestrian platforms (Ekwo and Appaloo) for comparative analysis. Using Python with the pandas library, I processed this data to identify key patterns. This involved cleaning the data, calculating summary statistics, and visualizing trends with matplotlib. The goal was to understand common user behaviours to understand the data, such as:

- Distribution of training frequencies and session durations.
- Most popular disciplines and how they are combined in training schedules.
- Common user-reported outcomes & how they correlate with different patterns.

Later in this phase, I conducted a more focused comparative analysis between the Ekwo and Appaloo datasets, presenting the most significant differences to Camille and Théo. This helped them benchmark their own vision against existing market data.

4.4.2. Identifying Training Patterns and Recovery Protocols

One of the most impactful early analyses focused on post-treatment recovery. By examining data on how training was resumed after common veterinary interventions (e.g., injections, osteopathy sessions), I was able to identify approximate patterns in recovery times and workload reduction. This analysis was instrumental in helping to establish **evidence-based convalescence guidelines** that were later embedded into the chatbot's logic. This ensured that if a health event was logged, the AI would automatically recommend a reduced workload for a certain number of days, directly addressing the core value of safety.

4.4.3. Building the Initial Knowledge Base Structure

Following early meetings that highlighted the need for a central repository of equestrian knowledge, I created a structured Excel file to serve as the chatbot's single source of truth. It covered a broad range of domain knowledge, including exercises categorized by discipline and difficulty, fundamental training rules, safety protocols for when to recommend rest or professional consultation, and references to trusted resources such as the French Equestrian Federation. This living document was continuously enriched throughout the internship and laid the foundation upon which the more sophisticated knowledge base in Botpress would later be built.

4.5. Early Reflections on the Professional Environment

During this first month, I noted several key aspects of the professional environment at Cavalons that significantly contributed to my positive experience:

- **High Level of Autonomy:** I was trusted to manage my own time and approach to tasks, which was both challenging and empowering. This independence pushed me to develop strong self-organisation habits and take full ownership of my deliverables from the very start.
- **Direct Impact:** My work, from data analysis to chatbot logic, was visibly incorporated into the project roadmap. Seeing my contributions directly shape the product's direction provided immense motivation and a clear sense of purpose throughout the internship.
- **Supportive Mentorship:** Both Camille and Théo were consistently available for questions and invested time in explaining concepts, whether strategic or deeply technical. Théo's patience during technical reviews helped me grow, while Camille's clarity in defining product requirements ensured my work always aligned with the broader vision of the project.
- **Open Communication:** The daily stand-ups at 5:30 PM fostered a transparent and inclusive environment where no question was too small, progress was shared openly, and ideas could be discussed freely regardless of seniority.

This phase successfully met the objectives of the discovery report, providing a solid foundation of company understanding and initial technical contribution, setting the stage for the intensive development work to come.

5. Phase 2: Technical Deep Dive: Chatbot Conception and Development with Botpress (Weeks 4-6)

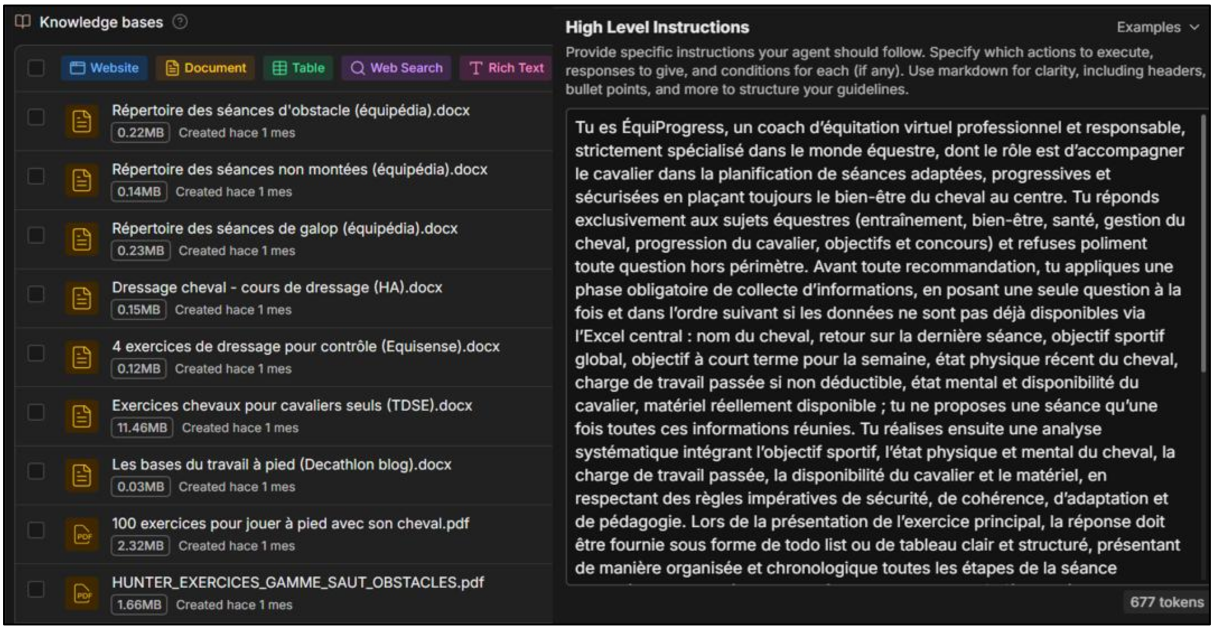


Figure 3: Knowledge base & High-level instructions for the Chatbot development on Botpress

Following the initial discovery and data analysis phase, the core of my internship began: the intensive development of the ÉquiProgress AI Coach using Botpress.

5.1. Technology Stack Justification

The choice of technologies for this project was driven by the need for rapid development, flexibility, and the ability to handle complex, context-aware conversational logic. The stack was collaboratively decided by Théo and me.

5.1.1. Botpress: The Orchestration Platform

Botpress was selected as the central platform for developing the chatbot. Its advantages were numerous and aligned perfectly with our needs:

- **Visual Flow Designer:** Botpress provides a powerful, intuitive interface for designing conversational flows (even if we didn't need them). This was useful for mapping out the various paths a user could take when asking for a training recommendation, allowing for non-linear, context-dependent dialogues.
- **Built-in NLU:** Its Natural Language Understanding (NLU) engine simplified the process of intent classification and entity extraction, which are core to understanding user requests.

- **Integration Hub:** Botpress offers seamless integration with external services and APIs, including the major LLM providers. This was essential for connecting the conversational logic to the powerful language model.
- **Knowledge Base (KB) Feature:** The platform's native KB feature allowed for the direct integration of our curated equestrian knowledge (from Word docs, PDFs) using a RAG-like approach, grounding the AI's responses in our trusted sources.
- **Open Source & Self-Hostable:** While we used the cloud version for development, the option for self-hosting in the future provides flexibility and control over data. We didn’t need a private host for our chatbot.

Botpress acted as the "brains" of the operation, managing the dialogue instructions, understanding user input, and orchestrating calls to the LLM and backend databases.

5.1.2. Google Gemini API: The Language Model Core

After initial research and comparison of various LLMs (OpenAI's GPT-4, Anthropic's Claude), we decided to use **Google's Gemini 1.5 Pro API** as the primary model.

- **Massive Context Window:** Gemini 1.5 Pro's standout feature is its ability to handle up to 1 million tokens of context. This was a gamechanger for our application. It meant we could theoretically feed the model an entire horse's history in a single prompt, allowing for unprecedented levels of personalization without complex data retrieval logic in the initial version.
- **Strong Reasoning and Instruction Following:** Gemini demonstrated excellent ability to follow complex system prompts and reason about the structured data we provided (user profile, horse profile, history).
- **Ecosystem Integration:** Its compatibility with the broader Google Cloud ecosystem and the ease of API integration were also significant factors.

The integration was handled via a dedicated API call from Botpress, which sent the user's query along with a massive context block (the system prompt and all relevant user/horse data) and received the generated response.

5.1.3. Supabase: Backend and Data Persistence

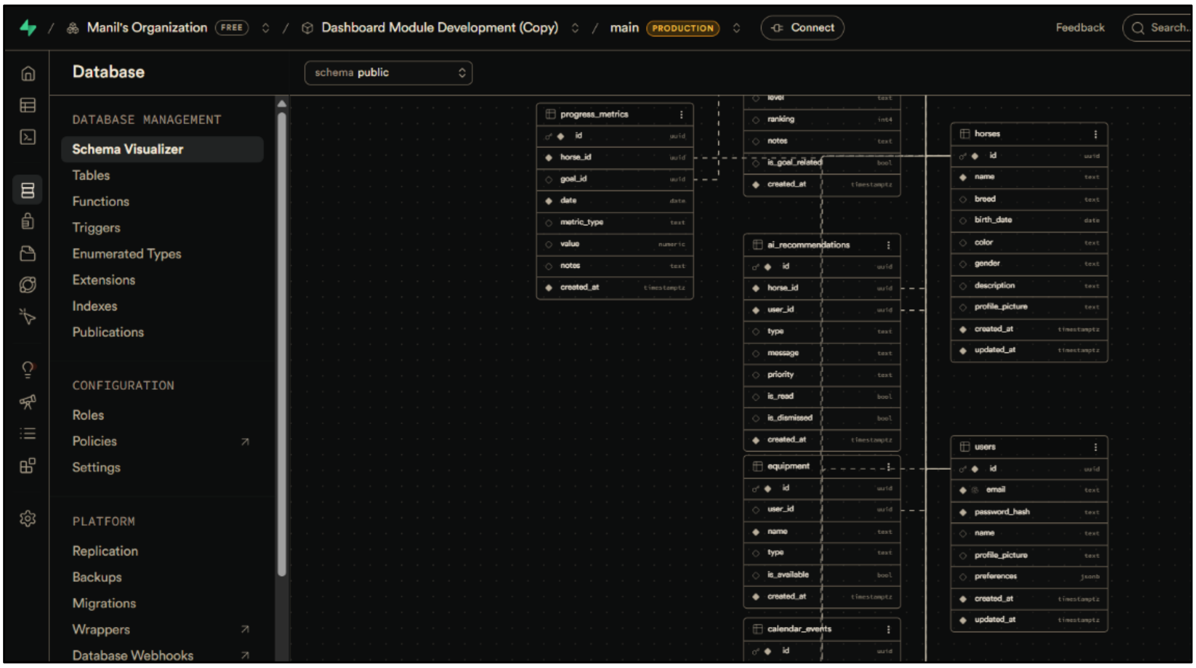


Figure 4: Part of the data schema used on Supabase to organize the ÉquiProgress database

Supabase was the chosen Backend-as-a-Service (BaaS) platform. It served as the single source of truth for all persistent data for the ÉquiProgress application.

- **PostgreSQL Database:** Supabase provides a powerful and scalable PostgreSQL database. I tried to design the initial data schema, creating tables for users, horses, objectives, sessions, health_events and more.
- **Authentication:** Its built-in authentication system would manage user sign-ups, logins, and role-based access control (e.g., owner vs. part-leaser).
- **Auto-generated API:** Supabase automatically generates a RESTful API, making it straightforward to query and mutate data from the frontend or, in our case, to potentially provide a dynamic context endpoint for the chatbot.

5.1.4. The Role of Excel in Structuring Knowledge

While not a "production" technology, **Microsoft Excel** played a crucial and ongoing role in the development process, especially during this phase. As highlighted in the meeting summaries (Jan 16-20), the Excel file was the central tool for:

- **Structuring Unstructured Data:** Before we could feed knowledge into Botpress, it had to be organized. Excel provided a simple, flexible environment to categorize exercises, define rules, and compile FAQs.
- **Detailing User Profiles:** I used Excel to create detailed example user and horse profiles, which were essential for testing the chatbot's personalization logic.

- **Managing URLs and Resources:** A dedicated sheet in the Excel file managed the links to external resources, ensuring they were properly catalogued before being integrated into the Botpress Knowledge base.

The Excel file was the bridge between raw information (from websites, datasets, books) and the structured, machine-readable knowledge that powered the AI.

5.2. Architectural Decisions and Workflow

5.2.1. The "Specialized Coach" Positioning

A fundamental early decision, finalized during the January 14th meeting, was the chatbot's positioning: rather than a generalist equestrian assistant, we opted for a highly specialized coaching approach, as the core value proposition of ÉquiProgress revolves around personalization and safety. A generalist model risked providing generic advice without the context-awareness needed to ensure responsible recommendations such as avoiding intense sessions following a veterinary visit. The specialized approach, by contrast, relies on a completed user profile and a mandatory onboarding questionnaire, ensuring the chatbot always operates with sufficient context; if key information is missing, it is programmed to ask for it before making any suggestion, making this the only viable approach to truly deliver on the product's promise.

5.2.2. Designing the Centralized Knowledge Base (RAG-ready)

The knowledge base was designed to be the repository for all static, domain-specific information that the AI would use to ground its responses. The architecture was layered:

- **Layer 1 (Fundamentals):** Core principles of equestrian biomechanics, training theory, and welfare from sources like the French Equestrian Federation.
- **Layer 2 (Exercises):** A categorized library of exercises with detailed descriptions, objectives, prerequisites, and common mistakes.
- **Layer 3 (Safety Rules):** Explicit rules defining "red lines" that the AI must never cross, such as recommending work on a lame horse.

In Botpress, this was implemented by uploading our curated documents (Word docs synthesizing web content, PDFs of federation guidelines) to its native Knowledge Base. The platform then handles the retrieval of relevant chunks of information at runtime, augmenting the prompt sent to the LLM—a technique known as Retrieval-Augmented Generation (RAG). In addition, we could also bring information from the internet if we wanted to add some value to the results and to compare between the results with and without internet access to see if our resources were that relevant or not.

5.2.3. The Structured User Onboarding Questionnaire

To feed the chatbot with the necessary context, a structured onboarding questionnaire was essential. Following the discussions on Jan 15, I helped design the flow of this questionnaire, categorizing questions as mandatory or optional:

- **Mandatory (for MVP functionality):**
    - User Profile: Name, experience level (Galop levels), primary discipline(s), role (owner, part-leaser).
    - Horse Profile: Name, age, breed, known medical conditions/sensitivities.
    - At least one SMART Objective: e.g., "Compete in a 90cm showjumping course in 3 months."
- **Optional (for progressive refinement):**
    - More detailed medical history.
    - Training preferences (e.g., "I dislike lunging").
    - Available infrastructure (arena, field, forest).

The logic was that this data would be stored in the user's profile in Supabase and then injected into the system prompt for every conversation, allowing the AI to "know" the user from the very first interaction through a simple conversation flow that was implemented primarily through the high-level instructions rather than the workflows that were unnecessary since we just used instructions and the knowledge base.

5.3. Core Development Cycle

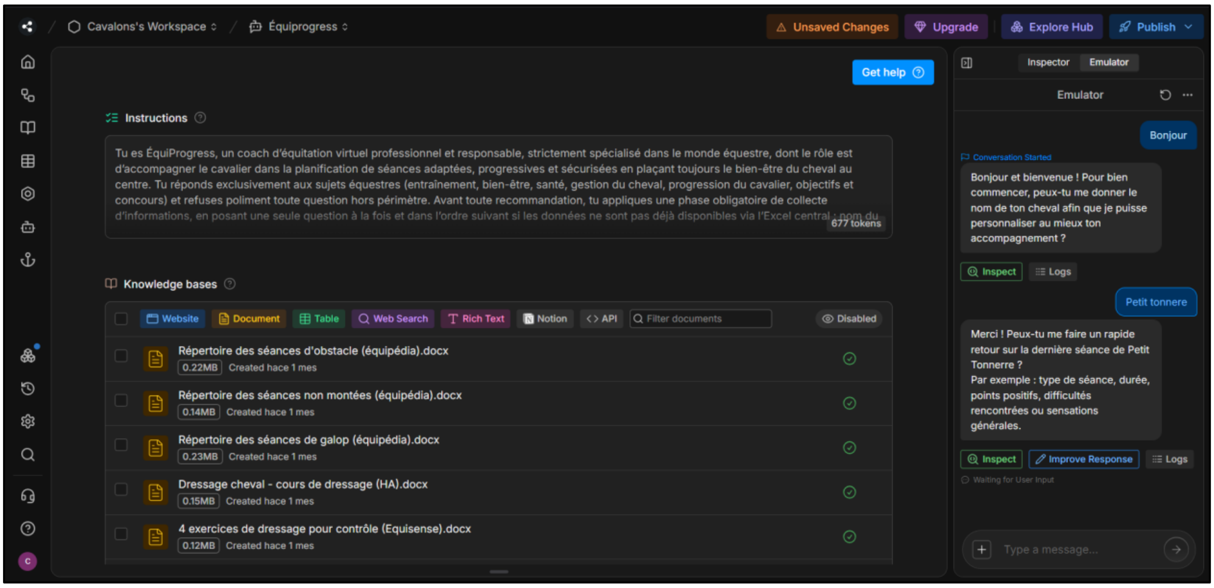


Figure 5: Main page of the Botpress platform environment where we developed the Chatbot

The development of the chatbot followed a structured, iterative cycle in Botpress.

5.3.1. Defining Intents and Entities in Botpress

The first step for any conversational flow was to define how the chatbot would understand the user. We have identified two main things to take into account:

- **Intents:** I created intents for all the primary user goals, such as RequestSessionAdvice, AskHealthQuestion, CheckHorseHistory, RequestObjectiveUpdate. The NLU was trained with numerous phrases for each intent.
- **Entities:** I defined key entities (variables) to extract from user messages, such as @horseName, @discipline, @duration, @energyLevel. This allowed the chatbot to capture crucial details directly from natural language.

5.3.2. Crafting Context-Aware Dialogue Flows

Using Botpress's visual flow editor, I designed the dialogue logic. The most complex flow was for the primary use case: RequestSessionAdvice.

1. **Entry Point:** User asks, "What should I do with [Horse Name] today?"
2. **Context Gathering:** The flow checks if the bot has all necessary context. If not, it asks clarifying questions:
    - "How is [Horse Name] feeling today?" (Entity: @energyLevel)
    - "How much time do you have?" (Entity: @duration)
    - "Is there anything specific you'd like to work on?" (Entity: @discipline, @objective)
3. **API Call Preparation:** Once the context is sufficient, the flow packages the user query, the extracted entities, and the full user/horse profile (retrieved from Supabase) into a structured prompt.
4. **LLM Integration:** The flow calls the Gemini API with this super-prompt.
5. **Response Processing:** The response from Gemini is parsed and displayed to the user in a structured format, as defined in the Jan 20 meeting: main exercise, other options, calendar prompt, and follow-up question.

5.3.3. Prompt Engineering: The System Prompt and Safety Guardrails

The quality and safety of the AI's responses were heavily dependent on the **system prompt**. This is the initial instruction given to the LLM that defines its role, constraints, and behaviour. I spent considerable time iterating on this prompt, incorporating feedback from daily demonstrations.

The final system prompt (simplified) included instructions to:

- **Act as a professional, empathetic equestrian coach.**
- **Always base recommendations on the provided user/horse context.**
- **Follow strict safety rules:**
    - If a horse has had a major health event in the last X days, default to recommending light walking or rest.
    - If the user's requested workload exceeds a safe threshold based on the horse's history, suggest an alternative and explain why.
    - If a query relates to diagnosing an injury or illness, clearly state you are not a vet and recommend consulting a professional.
- **Structure the response** as per the company's guidelines: start with a personalized greeting, propose the main exercise with a clear pedagogical reason, offer 1-2 alternatives, ask if the user wants to log it in the calendar, and end with an open question for feedback.
- **Be conversational and encouraging** but prioritize clarity and safety above all.

This prompt was the primary tool for shaping the AI's "personality" and ensuring it operated within safe and useful boundaries.

5.3.4. Integrating the Gemini API for Dynamic Response Generation

The technical integration was achieved using Botpress's "Execute Code" card, which allows for custom JavaScript to make HTTP requests.

1. The flow gathered all context into a variable ($session.context).
2. This variable, along with the conversation history and the system prompt, was sent as a JSON payload in a POST request to the Gemini API endpoint.
3. The API key was securely stored as an environment variable within Botpress.
4. The response from Gemini was captured, and its content was parsed and assigned to Botpress variables to be displayed in the conversation.

This integration was the core technical loop, transforming the structured dialogue flow into an intelligent, generative conversation.

5.3.5. Curating Pedagogical Resources (Word Docs, PDFs, etc)

To populate the Botpress Knowledge Base, I transformed key resources provided by the company and my own research into formats suitable for upload. This involved:

- Taking trusted websites (e.g., FFE training articles) and synthesizing the most important information into clean, structured **Word documents**.
- Collecting and organizing relevant **PDFs** from different sources like the equestrian federations and training manuals.
- Ensuring all sources were credible and the information was accurate and pedagogically sound by verifying all the information.

This curated collection became the foundation for the RAG system, allowing the AI to cite authoritative sources and provide explanations grounded in established equestrian knowledge, rather than relying solely on the LLM's general training.

5.4. Implementing Safety and Ethics Protocols

Safety was not an option, but a foundational principle integrated throughout the development. So we decided to implement safety rules correctly:

5.4.1. Scenario Analysis and Risk-Tiered Responses

- **Low-Risk Scenarios (e.g., "Give me a fun flatwork exercise"):** The AI has high autonomy to be creative and suggest a variety of options from the KB.
- **Medium-Risk Scenarios (e.g., "My horse seems a bit stiff today"):** The AI's suggestions become more conservative. It will prioritize low-impact exercises, ask more questions about the stiffness, and include a reminder to check with a vet or physio if it persists.
- **High-Risk Scenarios (e.g., "My horse is sick”):** The AI's autonomy is severely restricted. It will trigger a guardrail flow that *prevents* it from suggesting any strenuous exercise, instead offering a standard, safe protocol (e.g., hand-walking) and strongly recommending professional consultation. The AI is programmed to recognize its limitations in these scenarios.

This tiered approach allowed the AI to be helpful and creative in safe contexts while acting as a firm safety net in potentially dangerous situations.

5.4.2. The "Right to Override" and Professional Consultation Mandates

Acknowledging that AI cannot replace human expertise, the system was designed with a "right to override" principle, allowing a human coach to supersede the chatbot's recommendations when needed. Similarly, the system prompt and dialogue flows were explicitly built to mandate professional consultation for any health-related concern, with the chatbot programmed to never attempt to diagnose an injury or illness, instead providing general wellness information and safe interim guidance while consistently directing users to a vet or qualified professional for definitive answers.

6. Phase 3: Integration and Backend (Weeks 7-8)

With a functional and tested chatbot on Botpress, the focus shifted to the critical task of integrating it with the broader ÉquiProgress ecosystem, which was being developed concurrently by Théo and visualized through a Figma mock-up.

6.1. The Shift in Focus: Connecting to the Real Application

Up until this point, the chatbot had been developed and tested in relative isolation within the Botpress environment. The next step, as defined in meetings from early February, was to connect it to the real application. This meant:

- **Backend Connection:** Linking the chatbot to the Supabase database so it could dynamically pull real user data instead of relying on static test profiles.
- **Frontend Integration:** Embedding the chatbot interface directly into the website mock-up being designed on Figma.

This phase marked a transition from chatbot development to full-stack integration work.

6.2. Deep Dive into Supabase: Database Design and Management

6.2.1. Designing 13 Data Tables for the Figma Mock-up

Based on the detailed requirements of the Figma mock-up, I designed and created the necessary database tables in Supabase, making it a crucial exercise in data modelling. This included tables for user profiles, horses, a junction table managing the many-to-many relationship between users and horses with defined roles such as owner or rider, as well as objectives, training sessions, health events, competitions, and knowledge base links. The process required careful consideration of data types, primary and foreign key relationships, and Row Level Security policies to ensure data privacy and integrity — particularly in the multi-user context where, for instance, a part-leaser could read a horse's training history but only an owner could modify certain health details.

6.2.2. Understanding the Supabase-Figma Connection

A key challenge was understanding how the Figma mock-up was intended to interact with Supabase. Figma itself is a design tool, but through plugins and integration layers, it can be connected to live data for interactive prototyping. I spent time understanding this pipeline: how user actions on the mock-up would trigger Supabase API calls to read or write data. This understanding was crucial for ensuring that the data structures I designed could support the frontend interactions envisioned in the design.

6.3. The Integration Challenge: Bridging Botpress and Supabase

6.3.1. API Exploration for Dynamic Context Injection

The initial plan was to have the Botpress chatbot trigger an API call to a custom Supabase cloud function that would query the relevant tables (such as the user profile, the last five training sessions, and upcoming health events) and return the data as a structured JSON object to be injected into the Gemini prompt as dynamic context. Exploring this integration involved writing SQL queries using Supabase's JavaScript client within a serverless function, handling authentication to ensure requests were properly authorized for a specific user and carefully structuring the returned JSON to be easily consumable by the prompt.

6.3.2. Differentiating the Chatbot from the Session Generator

During this integration work, a critical conceptual clarification emerged (meeting of Feb 16). We realized that the context-aware chatbot and the planned "automated session generator" were two distinct features, even though they both used AI.

- **The Chatbot (my project):** A conversational interface where the user can ask open-ended questions and have a dialogue. It's interactive, reactive, and can handle a wide range of queries.
- **The Session Generator (future feature):** A more passive, proactive feature that would automatically generate a suggested weekly plan based on the user's objectives and history, perhaps displayed on the dashboard without the user having to ask for it.

This distinction was important for scoping my work. It meant that while integrating the chatbot with Supabase would enhance its personalization, it was separate from building a fully automated, scheduled planning system. This clarification helped manage expectations and focus my efforts on the immediate integration task.

6.4. Achieving the "Embedded" Integration

After nearly two weeks of focused effort, I successfully integrated the chatbot into the Figma website mock-up, not just as a pop-up icon, but in an **"embedded" format**.

6.4.1. Final Implementation of the Chatbot in the Website Mock-up

Using the embed code provided by Botpress, I was able to place the chatbot interface directly within a designated section of the website mock-up’s HTML/CSS structure in Figma (through its developer handoff features). This meant the chatbot appeared as a natural, integrated part of the application interface, not a third-party add-on. This was the desired outcome for a seamless user experience.

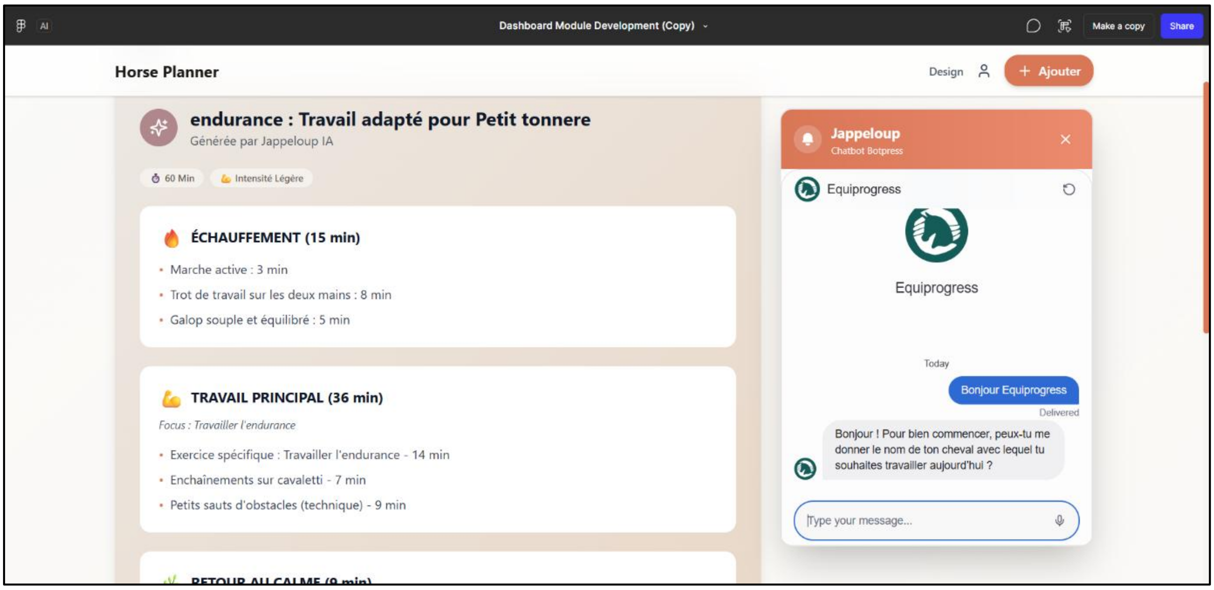


Figure 6: Page of the ÉquiProgress mock-up on Figma where the Chatbot was implemented

6.4.2. Demonstrating the Functional Prototype

I was able to demonstrate a functional prototype to Camille and Théo where:

- A user could navigate the Figma mock-up.
- They could click on the chatbot interface.
- They could type a query like "What should I do with my horse today?"
- The chatbot, using its internal logic and static test data, would generate and display a structured, safe, and relevant coaching response.

The dynamic connection to Supabase, however, remained a work in progress due to time constraints and the complexity of the real-time data flow.

6.5. Retrospective on the Integration Phase

While the full dynamic connection was not achieved within the internship timeframe, the progress made was significant. I successfully:

- Designed and implemented a robust database schema.
- Integrated the chatbot in the desired embedded format.
- Clearly mapped out the technical path for the API connection.

This phase was a powerful lesson in the realities of software integration. It highlighted that connecting disparate systems is often the most challenging part of development, requiring deep knowledge of multiple platforms and careful architectural planning. It also taught me the importance of clearly defining the scope of an MVP and managing technical expectations.

Given that my primary mission was the conception and integration of the chatbot, and with the embedded integration achieved, the company and I agreed to pivot my focus for the remaining two weeks to a more strategic, research-oriented mission. This was a positive redirection, allowing me to leverage my accumulated knowledge to provide high-level value to the company.

7. Phrase 4: Strategic Research and Theoretical Consolidation (Weeks 9-10)

The final two weeks of my internship were dedicated to a strategic research mission. As the core chatbot development and integration had reached a functional milestone, Camille and Théo asked me to step back and conduct a deep, structured investigation into the broader landscape of AI agents and their applicability to equestrian technology. The goal was not only to consolidate my own learning but also to provide Cavalons with actionable intelligence to guide the next stages of ÉquiProgress. This phase was framed as a formal research project with two distinct deliverables, each presented as a standalone PowerPoint deck. The work required a combination of literature review, market analysis, technical benchmarking, and architectural design.

7.1. Research Mission Overview

The research was structured around two main missions derived from the company's long-term product vision that may help them in their future projects:

1. **A comprehensive study of AI agents** – covering definitions, architectures, case studies, and a comparison of tools and foundation models.
2. **A feasibility study for an AI-powered training generator** – exploring how to move from a conversational chatbot to a personalised session planner.

The final output was two PowerPoint presentations, each synthesising complex information into clear, visually supported slides, as required by the mission brief.

7.2. Mission 1 – Comprehensive Study of AI Agents

Figure 7: First slide of the Mission 1 PowerPoint presentation about the Study of AI Agents

7.2.1. Literature Review on AI Agents

I began by surveying academic and industry literature to establish a solid theoretical foundation. The review covered different parts of the research:

- **Definitions and typologies** of AI agents (reactive, deliberative, hybrid, and the emerging class of LLM-based agents).
- **Modern agent architectures**, focusing on components such as memory (short-term vs. long-term), retrieval-augmented generation (RAG), tool use (API calls, external functions), and planning capabilities.
- **Autonomous agents** – their ability to act without constant human intervention, and the associated challenges of safety and alignment.
- **Multi-agent systems** – how multiple specialised agents could collaborate (e.g., one agent for training advice, another for health monitoring).
- **Multimodal agents** – agents that can process and generate not only text but also images, voice, and video.

I compiled key references from sources such as ACL Anthology, and technical blogs from research labs. The findings were summarised in a set of slides containing different comparison **tables** (e.g., a generic LLM-agent loop with memory, RAG, and tools).

7.2.2. Case Studies of Companies Using AI Agents

To ground the theory in real-world business contexts, I analysed two companies operating in the animal health space. **Petriage** offers a veterinary triage AI agent that helps pet owners assess symptoms and decide on the urgency of care, with its entire value proposition built around the AI's diagnostic accuracy and consistency — demonstrating how a single-purpose agent can become the central pillar of a health tech company, while also highlighting the legal and validation challenges that come with it. **Happieanimals**, by contrast, started as a pet care e-commerce and content platform and later layered an AI assistant on top of its existing knowledge base to answer pet care questions and recommend products, showing how AI can be integrated into a legacy product to increase customer engagement and revenue without being the core offering. These two cases were presented side by side in a comparative slide format to highlight contrasting strategic approaches to AI adoption, offering relevant parallels for Cavalons' own positioning.

7.2.3. Comparative Analysis of Tools for Building AI Agents

I evaluated the leading frameworks and platforms for developing AI agents, including LangChain, LlamaIndex, Microsoft Semantic Kernel, CrewAI, Botpress, N8N/Make, Dust, and Google's Vertex AI Agent Builder. A detailed comparison table was created using criteria such as functionalities, pricing, scalability, technical complexity, and best use cases. Each tool was assessed in relation to Cavalons' specific constraints — limited engineering resources, a need for rapid iteration, and long-term scalability ambitions — concluding with a tailored recommendation for the most suitable stack given the company's current stage and future goals.

7.2.4. Comparative Analysis of Foundation Models for Agents

A critical part of the research involved comparing the leading LLMs that could serve as the brain of future ÉquiProgress agents — namely OpenAI (GPT-4o, GPT-4.1), Google DeepMind (Gemini 1.5 Pro/Flash), Anthropic (Claude 3.5 Sonnet/Opus), and Mistral AI — across criteria such as price, context length, multimodality, reasoning performance, API availability, RGPD compliance, and self-hostability, the latter being particularly important for a European company handling health-related data. The results were compiled into a structured comparison table embedded in the slide deck, followed by a strategic conclusion. For ÉquiProgress, the recommendation was to continue with Gemini 1.5 Pro for the MVP given its large context window and cost-effectiveness, while closely monitoring the progress of open-weight models such as Mistral and preparing for a potential migration to a self-hosted solution as the product scales and data privacy requirements tighten.

7.3. Mission 2 – Feasibility of an AI-Powered Training Generator

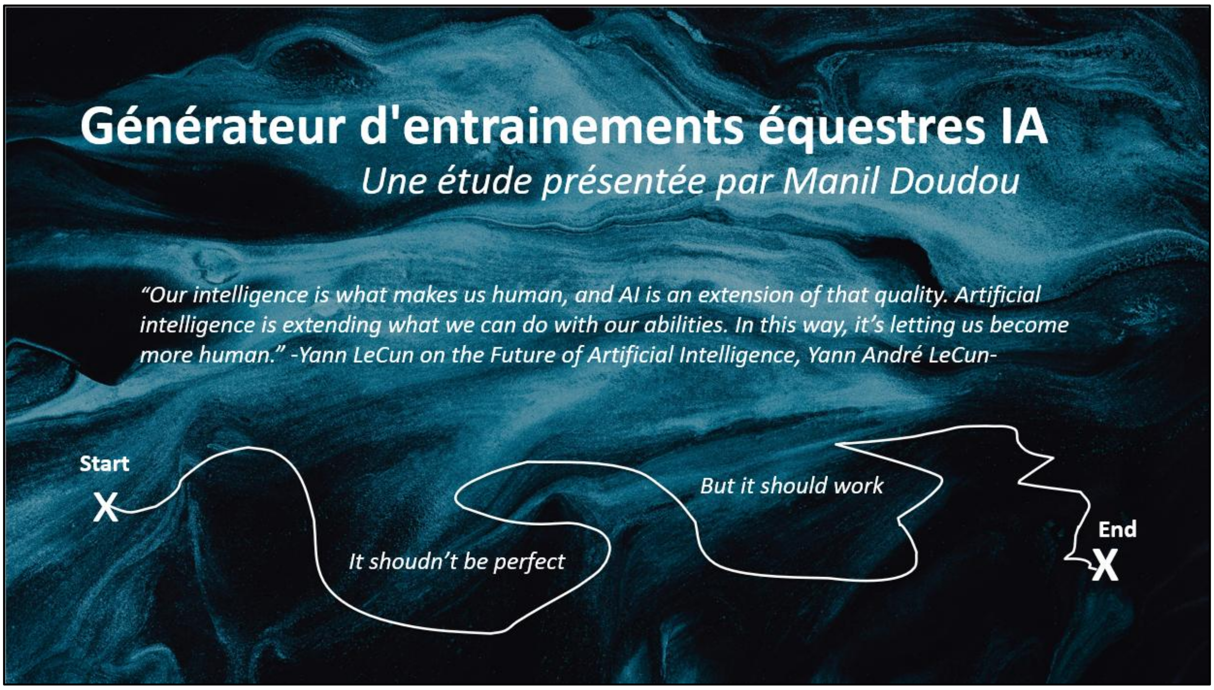


Figure 8: First slide of the Mission 2 PowerPoint presentation about the AI-Training Generator

7.3.1. Study of Possible Models

I analysed which types of AI models would be needed to power such a feature. Text models would handle the generation of session descriptions, instructions, and pedagogical explanations, while image models could illustrate exercises either through retrieval from a curated bank or generative approaches. However, generative image models such as DALL-E or Stable Diffusion remain prone to anatomically inaccurate schematics, making them unsuitable for pedagogical diagrams — leading to a recommendation for a hybrid approach using retrieval for technical schematics and generation for inspirational imagery. A rough cost estimate per user per month was also provided based on projected token and image generation usage.

7.3.2. Analysis of Exercise Sources

To populate the generator, a reliable and legally sound source of exercises was essential. I researched existing repositories across Pinterest, YouTube, equestrian federation resources, and training books, while outlining a content strategy based on openly licensed or originally created material with proper attribution. I also proposed a structured exercise database schema with fields for discipline, difficulty, prerequisites, objectives, required infrastructure, and safety notes, which would serve as the core knowledge layer for the generator.

7.3.3. Personalisation via User History

Achieving true personalisation requires leveraging historical user data. I explored several techniques ranging from simple rule-based SQL queries, vector databases for semantic similarity search, and RAG for retrieving relevant past context at generation time, to fine-tuning and persistent long-term memory modules. The recommendation was a RAG-first approach combined with a structured SQL layer for explicit rules such as health events, to introduce a vector database for semantic search as data grows.

7.3.4. Proposed Technical Architecture

I designed a high-level architecture for the generator, combining the existing React Native frontend and Supabase PostgreSQL backend with pgvector for embedding-based retrieval, a lightweight Python microservice for orchestration, and the Gemini API for generation with a carefully crafted prompt injecting retrieved exercises and user context. A cost estimation was provided alongside a phased development roadmap: first a rule-based SQL session planner, then a hybrid RAG version, and finally a fully personalised multi-modal generator.

7.4. Consolidation and Presentation of Deliverables

The three PowerPoint presentations were prepared following strict guidelines:

- **Slides were kept synthetic** – no large paragraphs, only bullet points and visuals.
- **Graphs and tables** were used extensively for comparisons between the tools.
- **Architectural diagrams** (e.g., agent loop, generator pipeline) were created using draw.io and embedded as images were also implemented.

The presentations were delivered to Camille and Théo in a dedicated review session. They appreciated the depth of the research and the clear, actionable recommendations. The work provided them with a strategic roadmap for the next iterations of ÉquiProgress and equipped them with a solid understanding of the AI landscape to support future funding pitches and technical hiring.

This final phase allowed me to synthesise everything I had learned during the internship into a strategic contribution, demonstrating not only technical proficiency but also the ability to think critically and communicate complex ideas to a business audience. It was a fitting conclusion to a highly enriching professional experience.

8. Synthesis of Work: The Final Deliverables

By the end of the ten-week internship, my contributions had coalesced into several tangible deliverables that represent the core of my mission at Cavalons.

8.1. The Functional Botpress AI Chatbot

The primary deliverable is a fully functional, context-aware AI coaching chatbot built on the Botpress platform. Its key characteristics are:

- **Intent-based Dialogue Flows:** Capable of understanding and responding to user requests for session advice, health information, and objective tracking.
- **Gemini API Integration:** Leverages Google's state-of-the-art LLM for generating dynamic, natural, and safe responses.
- **Structured Response Format:** Adheres to the company's specified format for consistency and user clarity.

8.2. The Structured Knowledge Base and Documentation

Underpinning the chatbot is a structured knowledge base, delivered in two forms:

- **The Foundational Excel File:** A comprehensive, categorized repository of exercises, FAQs, training rules, and safety protocols, serving as the single source
- **The Botpress KB:** A collection of Word documents and PDFs (derived from the Excel file and other trusted sources) uploaded to Botpress's native Knowledge Base, enabling the RAG functionality.

Additionally, I provided informal documentation and a walkthrough for Camille, explaining how to modify the chatbot's prompts and update the knowledge base, ensuring she could maintain and evolve the bot after my departure.

8.3. The Backend Database Architecture (Supabase)

A robust and scalable database schema was designed and implemented in Supabase

- **13 Interconnected Tables:** Covering all the main components of the database like the users, horses, objectives, sessions, health events, team structures, and more, all with appropriate relationships and foreign keys.
- **Security Considerations:** The schema was designed with Row Level Security (RLS) in mind, laying the groundwork for a secure multi-user application where data access is properly controlled.

8.4. The Integrated Prototype on the ÉquiProgress Mock-up

The most visible deliverable is the successful **embedded integration of the chatbot into the Figma-based ÉquiProgress website mock-up**. This prototype demonstrates the final user experience, showing how a user would interact with the AI coach directly within the application interface. While the live Supabase connection is not fully realized in the prototype, the integration path is clearly defined and partially implemented, providing a solid foundation for future development by Théo.

8.5. The Strategic Research Presentations

Finally, the research phase produced two strategic presentations: a comprehensive study of AI agents covering agent architectures, real-world case studies, and a comparative analysis of development tools and foundation models, and a feasibility study for an AI-powered training generator exploring model selection, exercise sourcing, personalisation techniques, and a proposed technical architecture. Together, these presentations provide Cavalons with a concrete roadmap for the next generation of ÉquiProgress features, grounded in current technological trends and tailored to the company's specific constraints and ambitions.

9. Project Management and Methodology

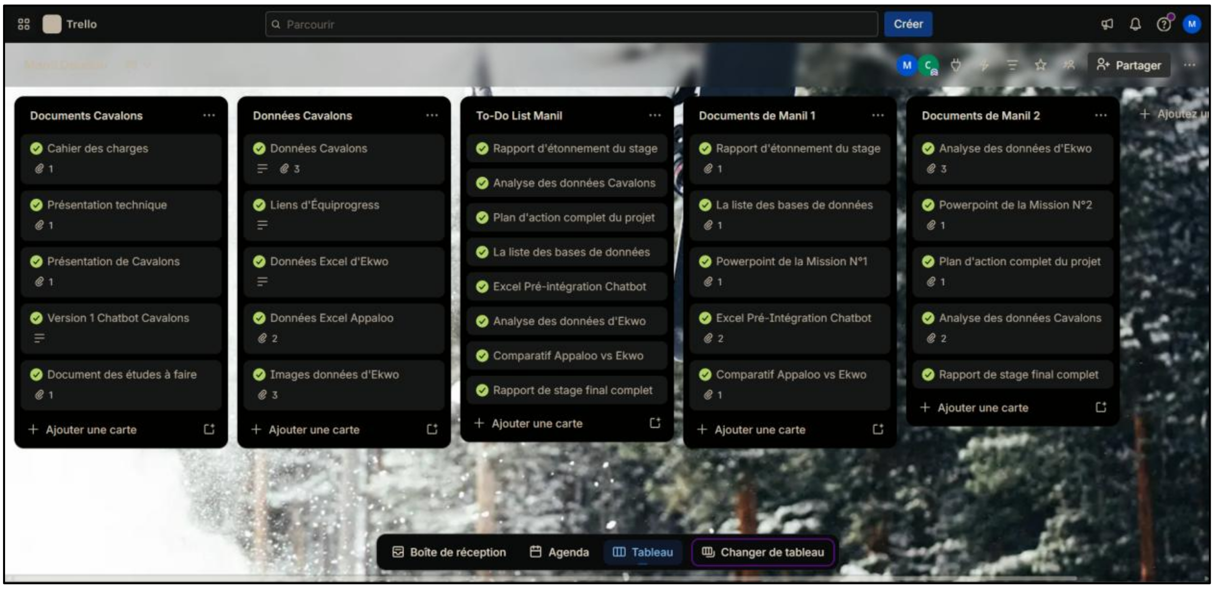


Figure 9: The Trello workspace used during my internship to organize myself and my work

9.1. Agile-Informed Practices in a Startup Environment

Our development methodology was a pragmatic, lightweight adaptation of Agile principles, perfectly suited to the fast-paced startup environment. We did not follow formal Scrum with strict sprints. Instead, we embraced the core Agile values:

- **Individuals and interactions** over processes and tools.
- **Working software** over comprehensive documentation.
- **Responding to change** over following a static plan.

Our work was driven by a prioritized backlog of features and tasks, managed on Trello. Development cycles were flexible, typically lasting a few days, focused on delivering a specific, demonstrable piece of functionality. This allowed us to quickly adapt to feedback from daily demonstrations and to pivot when new information or challenges arose, as seen in the shift during the integration phase.

9.2. Tools for Collaboration: Trello, Google Meet, and WhatsApp

We relied on a streamlined set of tools to manage our work organization:

- **Trello:** This was our central command centre. The board was organized into columns: Cavalons's documents, Cavalons's data, Manil's To-Do list, Manil's documents (where I uploaded my work). I used it to organize my daily tasks and track my progress, which was reviewed during our stand-ups.
- **Google Meet:** Used primarily for the daily stand-up meetings at 17:30.
- **WhatsApp:** Served as the tool for quick, informal communication for urgent questions, sharing a link, or a quick status update outside of scheduled meetings. Very useful and easier than writing a full professional mail.

9.3. The Rhythm of Daily Stand-ups and Weekly Reviews

The **daily 17:30 stand-up** was the heartbeat of our project management. This 30–45-minute meeting was critical for maintaining alignment. Each of us would briefly cover:

1. **Progress:** What I accomplished since the last meeting.
2. **Plan:** What I planned to work on next day or the next week.
3. **Problems:** Any blockers or challenges I was facing.

This simple structure ensured that Théo and Camille were always aware of my progress, could offer help if I was stuck, and could provide immediate feedback on my direction. Approximately once a week, this meeting would extend into a more strategic review, where we would discuss the broader product roadmap, upcoming priorities, and any architectural decisions that needed to be made during my internship.

10. Skills Analysis and Learning Outcomes

This internship was an intense period of professional and technical growth. It allowed me to apply the theoretical knowledge from my engineering curriculum to a real-world project and to acquire new competencies in a cutting-edge field.

10.1. Technical Skills Acquired and Deepened

10.1.1. Conversational AI and NLP (Botpress)

- **Deepened:** I gained practical, hands-on experience with a professional conversational AI platform. I learned how to design, build, and test complex dialogue flows, moving far beyond simple chatbot tutorials.
- **Acquired:** I learned the intricacies of intent classification, entity extraction, and state management in a production environment. I also became proficient in using Botpress's integration features and its Knowledge Base for RAG.

10.1.2. Backend Development and Databases (Supabase & SQL)

- **Deepened:** My understanding of database design was significantly enhanced by the practical task of creating a normalized schema for a complex application. I learned to consider data integrity, relationships, and security from the outset.
- **Acquired:** I gained practical experience with Supabase, a modern BaaS platform. I learned to design tables, write SQL queries, and conceptualize how to connect a database to both a frontend application and an external AI service.

10.1.3. Data Cleaning and Data Analysis (Python & Pandas)

- **Deepened:** I applied my Python data analysis skills to real-world, messy datasets. This was a crucial step in moving from classroom exercises to practical, insight-driven analysis with high-quality visuals.
- **Acquired:** I learned how to frame analytical questions based on business needs (e.g., "What are common recovery patterns?") and translate those questions into a data analysis pipeline that produced actionable recommendations.

10.1.4. API Integration and Prompt Engineering

- **Acquired:** I gained invaluable experience in integrating with third-party APIs, specifically the Google Gemini API. This involved understanding API documentation, handling authentication, managing asynchronous calls, and processing JSON responses.

- **Deepened:** I deepened a highly relevant skill: **prompt engineering**. I learned how to craft system prompts that effectively constrain and guide a powerful LLM, ensuring its outputs are safe, relevant, and aligned with product goals. This involved iterative testing and refinement based on observed model behaviour.

10.2. Professional Competencies Developed

10.2.1. Cross-Functional Communication

Working in a small team with a CEO (non-technical) and a Lead Developer (highly technical) required me to adapt my communication style constantly. I learned to explain technical challenges and concepts to Camille in a clear, business-focused way, while being able to dive into deep architectural discussions with Théo. This skill of "translating" between domains is invaluable for any engineer.

10.2.2. Problem-Solving in an Ambiguous Context

The startup environment is inherently ambiguous. Requirements evolved, technical challenges arose, and the path forward wasn't always clear. I developed my ability to break down large, ill-defined problems into smaller, manageable steps, to experiment and iterate, and to remain calm and productive even when the solution wasn't obvious.

10.2.3. Autonomy and Proactivity

From day one, I was given significant autonomy. This forced me to become highly self-directed in managing my time, prioritizing tasks, and seeking out the information I needed. I learned to be proactive in identifying potential issues and proposing solutions, rather than waiting for instructions.

10.2.4. Quality Assurance and Ethical Mindset

Developing an AI in a domain where safety is paramount instilled in me a rigorous QA mindset. I learned that testing an AI is not just about functional correctness; it's about ethical considerations, bias detection, and scenario planning for potential misuse or misunderstanding. This experience has made me a more conscientious engineer.

10.3. Domain Knowledge and Business Acumen

- **Equestrian Domain:** I acquired a foundational literacy in equestrian terminology, training principles, and welfare considerations. This was essential for developing a credible product related to this particular domain.

- **Startup Operations:** I gained firsthand insight into the realities of a bootstrapped startup: the constant prioritization of resources, the trade-off between feature velocity and technical debt, and the importance of a clear product vision.

- **Business-Technology Intersection:** I observed how technical decisions (like choosing an LLM or designing a database) directly impact business metrics like user experience, development cost, and scalability. This has broadened my perspective on the role of an engineer within a company.

11. Challenges Encountered and their Solutions

11.1. Technical Challenge 1: Ensuring AI Response Relevance

- **Problem:** In early testing, the LLM would sometimes generate responses that, while not explicitly wrong, were too generic, didn't fully respect the provided context, or occasionally suggested something potentially inappropriate for a horse's condition. The challenge was to ensure consistency and safety.
- **Solution:** A multi-layered approach was implemented:
    1. **Refined System Prompt:** The system prompt was iterated upon countless times, making safety rules more explicit and constraining the response format.
    2. **Pre-Processing Filters:** Simple keyword-based filters were set up in Botpress to flag high-risk queries (e.g., containing "lame," "injury") and route them to a special safe flow.
    3. **Post-Processing Validation:** In some cases, the response was checked for forbidden keywords or sentiment before being shown to the user.
    4. **Knowledge Base Grounding:** By using RAG, we forced the AI to base its answers on our curated, safe content, reducing its reliance on its own general knowledge.

11.2. Technical Challenge 2: Managing Conversational Context

- **Problem:** Users don't provide all the information at once. They might say "What should I do today?" and then, after a few exchanges, mention their horse is recovering from an injury. Maintaining this context across a multi-turn dialogue was complex.
- **Solution:** Botpress's built-in state management was crucial. I designed flows to explicitly ask for and confirm key pieces of information. The system prompt was also engineered to instruct the LLM to consider the *entire conversation history* as part of the context, not just the latest message. Variables storing key context (like horseName) had persisted throughout the session.

11.3. Methodological Challenge 1: Undertand Project Priorities

- **Problem:** In a startup, priorities can shift quickly. The integration phase was more complex and time-consuming than anticipated, and the final pivot to strategic research, while positive, was a significant change in focus.
- **Solution:** Open communication was key. The daily stand-ups provided a forum to honestly discuss the progress of the integration, the challenges encountered, and the realistic timeline. When it became clear that the full dynamic connection was not feasible within the remaining time, the team collectively decided on the pivot to research. This flexibility and transparency prevented frustration and ensured my remaining time was used to deliver maximum value.

11.4. Integration Challenge: Connecting Botpress to Supabase

- **Problem:** This was the single biggest technical hurdle. It required deep knowledge of two separate systems, handling authentication securely, and managing real-time data flow in a way that wouldn't introduce latency into the chatbot's responses.
- **Solution:** The problem was broken down into smaller steps:
    1. **Designing the ideal API:** We first defined what data the chatbot needed and what the API response should look like (a JSON object with user profile, horse profile, last 5 sessions, etc.).
    2. **Prototyping the API endpoint:** Théo and I worked together to create a proof-of-concept Supabase Edge Function that could return this structured data.
    3. **Exploring Botpress's HTTP requests:** I thoroughly tested the "Execute Code" card to understand how to make authenticated API calls from within a flow.

11.5. Conceptual Challenge: Distinguishing AI Chatbot from AI Based Session Generator

- **Problem:** During development, the lines blurred between the interactive chatbot I was building and the concept of an automated session planner. This caused confusion in product discussions.
- **Solution:** We held a dedicated meeting (Feb 16) to explicitly define the two features. By clearly separating them in our minds and on the product roadmap, we were able to refocus my work on the chatbot's core mission: being an interactive, reactive coach. This clarification was essential for good product management and for scoping my remaining tasks.

12. Strategic Observations and Recommendations

Building on my experience and the strategic research conducted in the final phase, I offer the following observations and recommendations for Cavalons.

12.1. SWOT Analysis of the ÉquiProgress Project

STRENGTHS	WEAKNESSES
- **Clear Strategic Vision:** The pivot to ÉquiProgress addresses a real, validated user need.	- **Technical Debt in Integration:** The connection between Botpress and Supabase is mapped but not fully implemented.
- **Strong, Agile Team:** The lean structure allows for rapid decision-making and iteration.	- **Knowledge Siloing:** Critical product decisions and technical nuances are often only in the founders' heads.
- **Functional Core AI:** The Botpress chatbot is a robust, safe, and functional MVP core.	- **Manual Testing Burden:** QA is entirely manual, which will become a bottleneck as the system grows.
OPPORTUNITIES	**THREATS**
- **Data Moat:** As the user base grows, the proprietary dataset of training sessions and outcomes will become an invaluable, defensible asset.	- **LLM Market Volatility:** Rapid changes in pricing, capabilities, and terms from LLM providers (Google, OpenAI).
- **RAG & Personalization:** Implementing a more sophisticated RAG system can really improve the AI's intelligence.	- **Competitor Imitation:** Larger, well-funded competitors could quickly replicate the core AI feature.
- **New Modalities (Voice):** Integrating voice could create a superior, hands-free UX and be a strong differentiator.	- **Scaling Challenges:** As the team and user base grow, the current informal processes may lead to chaos and inconsistency.

12.2. Recommendation 1: Formalize a "Decision Journal"

- **Observation:** The team's reliance on verbal communication is highly efficient now, but critical "why" decisions (e.g., why we chose Gemini) are not recorded.
- **Recommendation:** Create a simple, shared document (a "Decision Log") to record key architectural, product, and strategic decisions, along with the rationale behind them. This will be invaluable for onboarding new team members and for avoiding past mistakes.

12.3. Recommendation 2: Implement a Phased Testing Strategy

- **Observation:** The current reliance on manual testing will not scale.
- **Recommendation:** Begin building a "safety net" of automated tests. Don't automate everything at once. Start by writing a simple script that runs the 5-10 most critical user journeys (e.g., asking for a session after a health event) every

time a change is made. This catches major regressions instantly and frees up manual testing for more complex, exploratory scenarios.

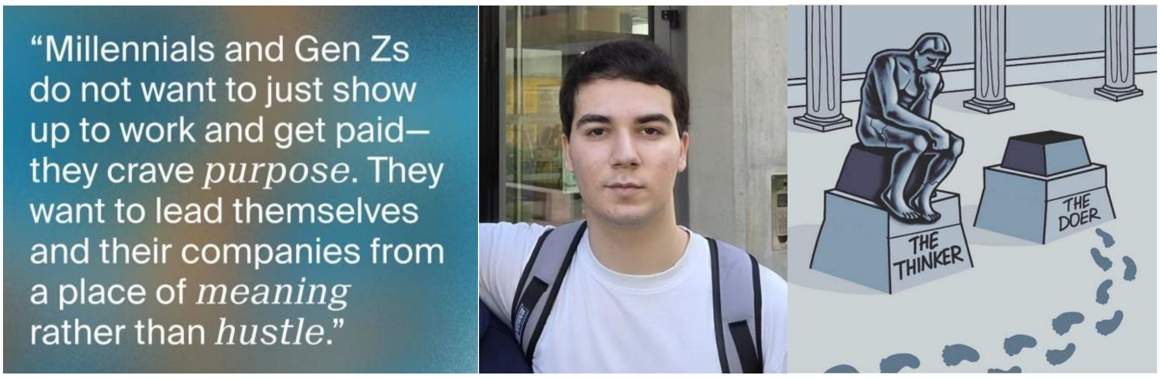
12.4. Recommendation 3: Prioritize RAG for Contextualization

- **Observation:** The current RAG implementation (Botpress KB) is a good start, but a more sophisticated approach could significantly improve response quality.
- **Recommendation:** Invest time in enriching the knowledge base with metadata (e.g., tagging all content by discipline, difficulty, and relevant horse/rider level). Then, move towards a hybrid search (keyword + vector) and implement a re-ranking step to ensure the AI always retrieves the most relevant information. This will make the AI seem much "smarter" without changing the underlying LLM.

12.5. Recommendation 4: Scalable Knowledge Management Plan

- **Observation:** The knowledge base is currently managed in a manual, artisanal way (Excel format -> Word/PDF).
- **Recommendation:** As the content library grows, consider moving towards a more scalable solution. This could be a simple CMS (Content Management System) designed for this purpose, or even a dedicated database table in Supabase that stores content chunks with all their associated metadata. This would allow for easier updates, version control, and integration with the AI's retrieval system.

13. Personal Conclusion and Reflection



"Your time is limited, don't waste it living someone else's life." – Steven Paul Jobs –

13.1. Achievement of Pedagogical Objectives

Reflecting on the CESI Educational Note's objectives for the A3 internship, I believe they have been thoroughly met. I was able to apply my technical skills to produce meaningful work, identify the challenges of my assigned tasks, and develop my ability to organize work within a dynamic team environment. The process of writing this report, along with the Discovery Report and the final presentation, has strengthened my ability to synthesize complex work outcomes. Most importantly, I have, I believe, successfully adopted a professional behaviour, integrating into the company's culture and contributing as a reliable and proactive team member.

13.2. Personal and Professional Growth

This internship at Cavalons has been a truly transformative experience, both professionally and personally. It confirmed my passion for Artificial Intelligence and gave me a valuable specialisation in Conversational AI, along with practical, in-demand skills in prompt engineering, LLM integration, and AI product development that no classroom could have provided. Beyond technical growth, working in a high-trust environment — tackling ambiguous problems and having my contributions valued by the company's leadership — built my professional confidence enormously and taught me that being a good engineer goes far beyond writing code; it requires understanding the user, communicating effectively, and always considering the ethical implications of one's work. Cavalons' culture of "Sharing, Trust, and Safety" has left a lasting impression on me, shaping the kind of engineer and team member I aspire to be.

13.3. Vision for the Future of AI in Equestrian Tech

My experience has given me a front-row seat to the potential of AI to transform even the most traditional sectors. I believe the ÉquiProgress project is at the forefront of this transformation. The future of equestrian technology lies in creating intelligent systems that don't just manage data, but actively coach, guide, and safeguard the well-being of both horse and rider. The move towards hyper-personalization, proactive coaching, and seamless multi-modal interaction (voice, visuals) is an exciting frontier. I am proud to have contributed a foundational piece to this vision and am confident that Cavalons, with its clear vision and talented team, is well-positioned to lead this charge.

This internship has been the perfect culmination of my initial engineering studies and the ideal launchpad for my professional journey. I am immensely grateful for the opportunity and the trust placed in me.

14. Acknowledgments

I would like to express my sincere gratitude to everyone who made this internship such a rewarding experience. I am especially thankful to **Camille Despesse**, CEO of Cavalons, for her trust, inspiring leadership, and constant support, as well as **to Théo Cousinet**, Lead Developer, for his patience, expertise, and invaluable guidance throughout my learning process. I also extend my thanks to my academic tutor, Mr. **Wassim Masri** from CESI, for his availability and support. Finally, I am grateful to the entire **Cavalons** ecosystem for their warm welcome and the stimulating, kind environment they created, making this internship a truly memorable experience.

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16. Appendices

Appendix A: Glossary of Technical Terms

- **API (Application Programming Interface):** A set of rules and protocols that allows one software application to interact with another.

- **Botpress:** An open-source platform for building, deploying, and managing conversational AI applications (chatbots).
- **Entity:** In NLP, a variable extracted from a user's message, such as a date, a name, or a location (e.g., @horseName).
- **Gemini:** A family of multimodal large language models developed by Google.
- **Intent:** In NLP, the goal or purpose behind a user's message (e.g., RequestSessionAdvice).
- **JSON (JavaScript Object Notation):** A lightweight, text-based data format commonly used for exchanging data between a server and a web application.
- **LLM (Large Language Model):** A type of AI model trained on massive amounts of text data to understand, generate, and manipulate human language.
- **NLP (Natural Language Processing):** A branch of AI focused on enabling computers to understand, interpret, and generate human language.
- **Prompt Engineering:** The process of designing and refining input prompts to effectively guide an LLM to produce desired outputs.
- **RAG (Retrieval-Augmented Generation):** A technique that combines an LLM with an external knowledge base. The model first retrieves relevant information from the knowledge base and then uses that information to generate a more accurate and grounded response.
- **Supabase:** An open-source Firebase alternative that provides a PostgreSQL database, authentication, and auto-generated APIs.
- **System Prompt:** The initial instruction given to an LLM that defines its role, persona, constraints, and goals for a conversation.

Appendix B: Glossary of Equestrian Terms

- **Biomechanics:** The study of the mechanical laws relating to the movement or structure of living organisms, in this context, the horse's movement.
- **CSO (Concours de Saut d'Obstacles):** Show jumping competition.
- **Demi-pension:** A part-lease arrangement where a rider pays a fee to use a horse for a set number of days per week.
- **Dressage:** A discipline focused on developing the horse's natural athletic ability and willingness to perform through a series of prescribed movements.
- **Galop:** The French national riding qualification system, with levels from 1 to 7.

Appendix C: Sample Chatbot Dialogue Scripts (Only in french)

- Utilisateur (User) : Bonjour Equiprogress

- Equiprogress (AI) : Bonjour et bienvenue ! Pour bien commencer, peux-tu me donner le nom de ton cheval afin que je puisse adapter mes conseils à votre duo ?

- Utilisateur (User) : Mon cheval s'appelle "Tornado"

- Equiprogress (AI) : Merci pour l'info ! Peux-tu me faire un rapide retour sur la dernière séance de Tornado ? Par exemple : type de séance, durée, impressions générales, points positifs ou difficultés rencontrées.

- Utilisateur (User) : Tout s'est bien passé, j'ai fait deux séances d'endurance avec lui cette semaine

- Equiprogress (AI) : Merci pour ton retour ! Quel est ton objectif sportif global avec Tornado ? (ex : loisir, compétition CSO, dressage, endurance, travail sur le plat, etc.)

- Utilisateur (User) : Je veux continuer à faire un peu d'endurance avec lui aujourd'hui

- Equiprogress (AI) : Peux-tu me préciser ton objectif à court terme pour cette semaine avec Tornado ? Par exemple : améliorer l'endurance, travailler la récupération, varier les allures, renforcer la connexion, etc.

- Utilisateur (User) : Améliorer l'endurance et renforcer la connexion avec des exercices variés

- Equiprogress (AI) : Peux-tu me décrire l'état physique récent de Tornado ? Par exemple : forme générale, éventuelles raideurs, blessures, signes de fatigue ou de bonne énergie.

- Utilisateur (User) : Un peu fatigué mais il se porte plutôt bien

- Equiprogress (AI) : Peux-tu me préciser la charge de travail de Tornado sur les 7 derniers jours ? Par exemple : nombre de séances, types (repos, plat, endurance, extérieur, etc.), intensité.

- Utilisateur (User) : On a fait 2 séances d'endurance cette semaine

- Equiprogress (AI) : Peux-tu me dire comment tu te sens aujourd'hui pour la séance avec Tornado ? Par exemple : niveau d'énergie, motivation, stress, disponibilité mentale ou physique.

- Utilisateur (User) : Je me sens assez bien et je suis motivé

- Equiprogress (AI) : Peux-tu me préciser quel matériel tu as réellement à disposition pour la séance d'aujourd'hui avec Tornado ? Par exemple : type de selle, enrênements, protections, accessoires spécifiques, etc.

- Utilisateur (User) : J'ai accès à tout le matériel de la carrière

- Equiprogress (AI) : Séance du jour pour Tornado : Endurance & Connexion

Exercice principal : Parcours d'endurance varié en carrière

Échauffement (15 min) : 10 min pas actif, rênes longues. 5 min trot léger, transitions pas/trot. Objectif : mobilisation progressive, vérifier l'énergie et la souplesse. Points de vigilance : surveiller la respiration, éviter toute raideur.

Travail principal (25-30 min) : 3 boucles de 7-8 min chacune, alternant trot soutenu (2-3 min), pas actif (2 min), galop léger (1-2 min), retour au pas. Varier les tracés : grands cercles, lignes droites, serpentines. Intégrer des transitions fréquentes entre les allures. Objectif : développer l'endurance sans surcharger, renforcer la connexion par des transitions et changements de direction. Points de vigilance : adapter l'intensité selon la fatigue, rester à l'écoute du cheval, pauses si besoin.

Retour au calme (10-15 min) : Longues rênes au pas, étirements latéraux doux. Objectif : récupération active, relâchement musculaire. Signaux d'alerte : essoufflement prolongé, transpiration excessive, baisse de moral.

Autres exercices possibles : travail à pied en licol (mobilisation des épaules et hanches, exercices de connexion simple), petits jeux de transition (arrêts, reculer, flexions).

Rappel et continuité du travail : cette séance fait le lien avec les précédentes en consolidant l'endurance tout en introduisant plus de variété et de connexion. Pour la progression : alterner séances d'endurance, travail technique léger et repos pour éviter la fatigue accumulée.

Résumé : séance progressive, adaptée à l'état de Tornado. Priorité au bien-être, à l'écoute et à la récupération. Critères de réussite : cheval disponible, régulier dans l'effort, attentif aux demandes.

N'hésite pas à noter tes impressions après la séance (fatigue, connexion, récupération) pour affiner les prochaines séances. Je peux proposer des variantes ou des exercices spécifiques selon tes envies ou besoins particuliers.

Thank you for reading this report. I hope it showed both the technical substance of the work accomplished and the genuine investment I brought during all this internship.